Math Practice Worksheet

Pythagorean Theorem · Absolute Value Graphs · Evaluate Functions · Simplify Roots

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1 Pythagorean Theorem

Key Formula

In a right triangle with legs a, b and hypotenuse c (longest side): $\boldsymbol{a^2 + b^2 = c^2}$
Find missing side: isolate the unknown and take the square root.
Check right triangle: substitute all three sides — if $a^2 + b^2 = c^2$ then **Yes**, otherwise **No**.

Example

(a) Find hypotenuse: $a = 3,\ b = 4$ $\quad c^2 = 3^2 + 4^2 = 9 + 16 = 25 \ \Rightarrow\ c = \sqrt{25} = \boxed{5}$

(b) Find missing leg: $a = 5,\ c = 13$ $\quad b^2 = 13^2 - 5^2 = 169 - 25 = 144 \ \Rightarrow\ b = \sqrt{144} = \boxed{12}$

(c) Check $(3, 4, 5)$**:** $3^2 + 4^2 = 9 + 16 = 25 = 5^2$ ✓ $\quad \boxed{\text{Yes, right triangle}}$

1. Find the hypotenuse: $a = 6,\ b = 8$.

2. Find the hypotenuse: $a = 5,\ b = 12$.

3. Find the hypotenuse: $a = 7,\ b = 24$.

4. Find the hypotenuse: $a = 10,\ b = 24$.

5. Find the missing leg: $a = 9,\ c = 15$.

6. Find the missing leg: $b = 8,\ c = 17$.

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7. Find the missing leg: $a = 20,\ c = 25$.

8. Is (6, 8, 10) a right triangle?

9. Is (5, 7, 9) a right triangle?

10. Is (8, 15, 17) a right triangle?

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2 Graphing Absolute Value Functions

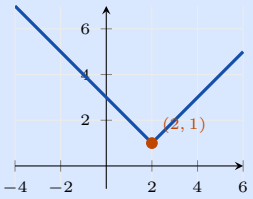
Transformation Summary Table: $y = a|x - h| + k$

Form	Transformation	Effect on Graph
$y = \|x\|$	Parent function	V-shape, vertex at $(0, 0)$, opens up
$y = \|x\| + k$, $k > 0$	Vertical shift **up**	Vertex moves up k units
$y = \|x\| - k$, $k > 0$	Vertical shift **down**	Vertex moves down k units
$y = \|x - h\|$, $h > 0$	Horizontal shift **right**	Vertex moves right h units
$y = \|x + h\|$, $h > 0$	Horizontal shift **left**	Vertex moves left h units
$y = a\|x\|$, $a > 1$	Vertical **stretch**	V becomes **narrower**
$y = a\|x\|$, $0 < a < 1$	Vertical **compression**	V becomes **wider**
$y = -\|x\|$	**Reflection** over x-axis	Opens **downward**
$y = a\|x - h\| + k$	**Combined** transformation	Vertex (h, k); up if $a > 0$, down if $a < 0$

Key Concepts

Vertex $= (h, k)$ $a > 0$: opens **up** $a < 0$: opens **down** $|a| > 1$: **narrower** $0 < |a| < 1$: **wider**

Example: Graph $y = |x - 2| + 1$



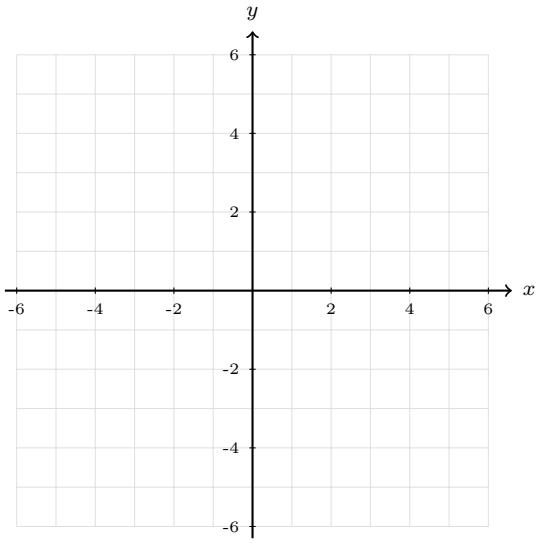
Vertex $(2, 1)$; $a = 1 > 0$ opens up; slopes ± 1 from vertex.

Graph each function. Label the vertex and state the transformation(s).

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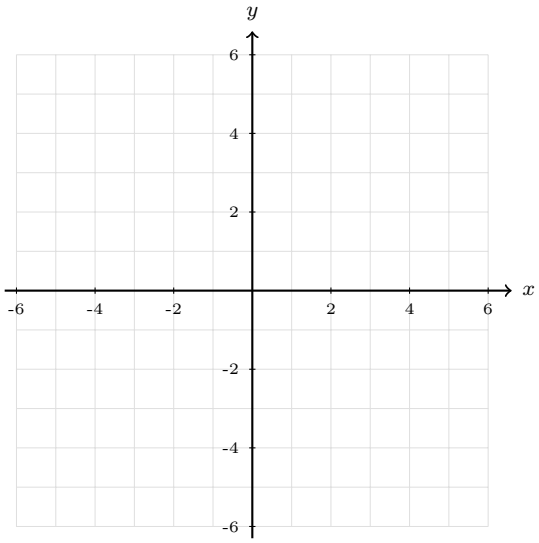
1. $y = |x|$

Vertex:_____ Trans:_____



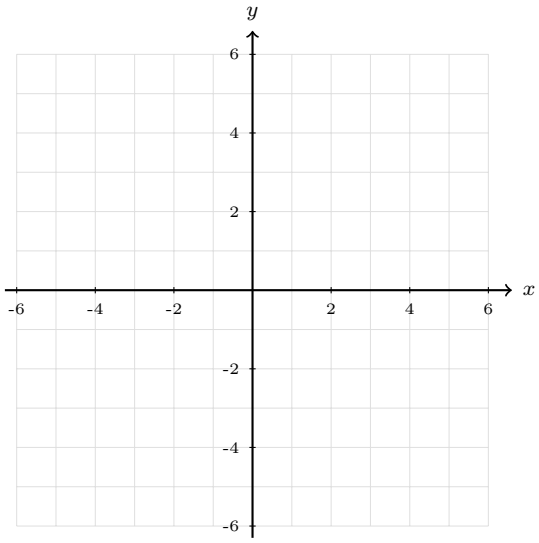
2. $y = |x| + 3$

Vertex:_____ Trans:_____



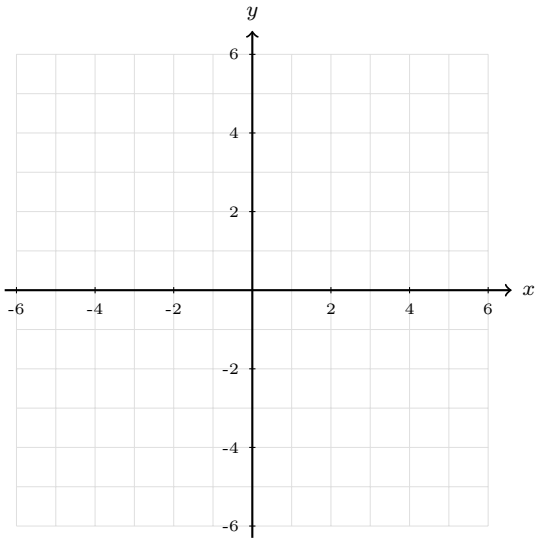
3. $y = |x| - 4$

Vertex:_____ Trans:_____



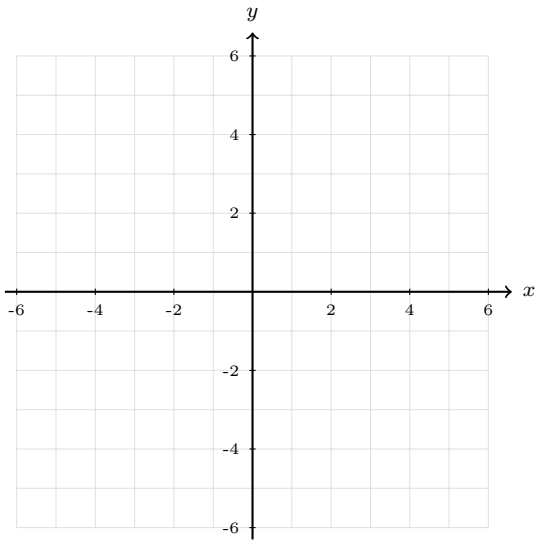
4. $y = |x - 3|$

Vertex:_____ Trans:_____



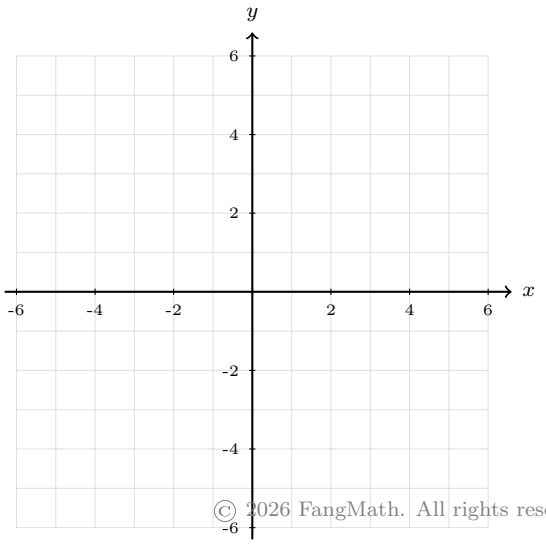
5. $y = |x + 2|$

Vertex:_____ Trans:_____



6. $y = |x - 1| + 2$

Vertex:_____ Trans:_____



7. $y = |x + 3| - 2$

8. $y = 2|x|$

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3 Evaluate Functions

Key Concept

To evaluate $f(a)$: replace every x with a, then simplify carefully (watch signs!).

Example

Let $f(x) = 3x^2 - 2x + 1$. Find $f(-2)$.

$f(-2) = 3\underbrace{(-2)^2}_{=4} - 2(-2) + 1 = 12 + 4 + 1 = \boxed{17}$

1. $f(x) = 2x + 5$. Find $f(4)$.

2. $g(x) = x^2 - 3$. Find $g(-2)$.

3. $h(x) = 3x^2 - x + 2$. Find $h(3)$.

4. $f(x) = |x - 5|$. Find $f(1)$.

5. $g(x) = \sqrt{x + 7}$. Find $g(9)$.

6. $f(x) = 2x^3 - 4$. Find $f(-1)$.

7. $h(x) = \dfrac{x + 3}{x - 1}$. Find $h(5)$.

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8. $f(x) = 5x - x^2$. Find $f(-3)$.

9. $g(x) = x^2 + 4x - 7$. Find $g(-4)$.

10. $h(x) = 2|x + 1| - 3$. Find $h(-5)$.

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4 Simplify Radical Expressions (Hard Level)

Key Rules

$\sqrt{a^n} = a^{n/2}$ $\sqrt[3]{a^n} = a^{n/3}$ $\sqrt{ab} = \sqrt{a} \cdot \sqrt{b}$ $\sqrt[3]{ab} = \sqrt[3]{a} \cdot \sqrt[3]{b}$

Add/Subtract like radicals only: $m\sqrt{a} \pm n\sqrt{a} = (m \pm n)\sqrt{a}$ $m\sqrt[3]{a} \pm n\sqrt[3]{a} = (m \pm n)\sqrt[3]{a}$

Examples

(a) $3\sqrt{12} + 2\sqrt{27} = 6\sqrt{3} + 6\sqrt{3} = \boxed{12\sqrt{3}}$ **(b)** $\sqrt{50} - \sqrt{8} = 5\sqrt{2} - 2\sqrt{2} = \boxed{3\sqrt{2}}$

(c) $2\sqrt[3]{16} + \sqrt[3]{54} = 4\sqrt[3]{2} + 3\sqrt[3]{2} = \boxed{7\sqrt[3]{2}}$ **(d)** $\sqrt{\dfrac{a^8b^6}{a^2b^2}} = \sqrt{a^6b^4} = \boxed{a^3b^2}$

1. $5\sqrt{12} + 3\sqrt{75}$

2. $\sqrt{98} - 2\sqrt{50} + \sqrt{32}$

3. $3\sqrt[3]{16} + 2\sqrt[3]{54}$

4. $4\sqrt[3]{81} - 2\sqrt[3]{24}$

5. $\sqrt{48x^5}$

6. $\sqrt[3]{x^{12}y^9}$

7. $\sqrt{\dfrac{a^8b^6}{a^2b^2}}$

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8. $2\sqrt{45x^5} - x^2\sqrt{20x}$

9. $3\sqrt{8x^3} - x\sqrt{18x}$

10. $\sqrt[3]{24a^6b^3}$

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5 Extra Challenge: Simplify Radicals (Advanced)

Strategies

Factor perfect squares/cubes from numbers and variables first, then combine like radicals.

$\sqrt[3]{A} \cdot \sqrt[3]{B} = \sqrt[3]{AB}$ $\quad \dfrac{\sqrt{A}}{\sqrt{B}} = \sqrt{\dfrac{A}{B}}$

1. $5\sqrt{18x^3} + 2x\sqrt{50x} - \sqrt{2x^3}$

2. $\sqrt[3]{54x^4y^7}$

3. $2\sqrt[3]{40} - 3\sqrt[3]{135}$

4. $\sqrt[3]{16a^7b^4} + 2ab\sqrt[3]{2a^4b}$

5. $\dfrac{\sqrt{72a^5b^3}}{\sqrt{2ab}}$

6. $\sqrt[3]{\dfrac{x^9y^3}{8z^6}}$

7. $4\sqrt{75x^3} - x\sqrt{48x} + \sqrt{27x^3}$

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8. $\sqrt[3]{x^5y^4} \cdot \sqrt[3]{x^4y^2}$

9. $3\sqrt{a^3b^5} - ab\sqrt{ab^3}$

10. $\sqrt[3]{250} - \sqrt[3]{2} + \sqrt[3]{16}$

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— Answer Key —

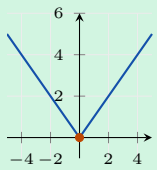
Part 1 — Pythagorean Theorem

1. $c^2 = 36 + 64 = 100 \Rightarrow c = \mathbf{10}$
2. $c^2 = 25 + 144 = 169 \Rightarrow c = \mathbf{13}$
3. $c^2 = 49 + 576 = 625 \Rightarrow c = \mathbf{25}$
4. $c^2 = 100 + 576 = 676 \Rightarrow c = \mathbf{26}$
5. $b^2 = 225 - 81 = 144 \Rightarrow b = \mathbf{12}$
6. $a^2 = 289 - 64 = 225 \Rightarrow a = \mathbf{15}$
7. $b^2 = 625 - 400 = 225 \Rightarrow b = \mathbf{15}$
8. $36 + 64 = 100 = 10^2$ ✓ **Yes**
9. $25 + 49 = 74 \neq 81$ **No**
10. $64 + 225 = 289 = 17^2$ ✓ **Yes**

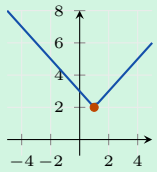
Part 2 — Graphing Absolute Value

1. Vertex $(0, 0)$; parent
2. Vertex $(0, 3)$; up 3
3. Vertex $(0, -4)$; down 4
4. Vertex $(3, 0)$; right 3
5. Vertex $(-2, 0)$; left 2
6. Vertex $(1, 2)$; right 1, up 2
7. Vertex $(-3, -2)$; left 3, down 2
8. Vertex $(0, 0)$; narrower
9. Vertex $(0, 0)$; opens down
10. Vertex $(-2, 4)$; left 2, up 4, opens down

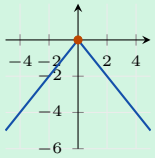
1. $y = |x|$



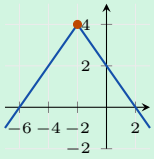
6. $y = |x - 1| + 2$



9. $y = -|x|$



10. $y = -|x + 2| + 4$



Part 3 — Evaluate Functions

1. $f(4) = 8 + 5 = \mathbf{13}$
2. $g(-2) = 4 - 3 = \mathbf{1}$
3. $h(3) = 27 - 3 + 2 = \mathbf{26}$
4. $f(1) = |1 - 5| = \mathbf{4}$
5. $g(9) = \sqrt{16} = \mathbf{4}$
6. $f(-1) = -2 - 4 = \mathbf{-6}$
7. $h(5) = 8/4 = \mathbf{2}$
8. $f(-3) = -15 - 9 = \mathbf{-24}$
9. $g(-4) = 16 - 16 - 7 = \mathbf{-7}$

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10. $h(-5) = 2(4) - 3 = \mathbf{5}$

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Part 4 — Simplify Radicals (1–5)

1. $5\sqrt{12}+3\sqrt{75}=10\sqrt{3}+15\sqrt{3}=\mathbf{25\sqrt{3}}$
2. $\sqrt{98}-2\sqrt{50}+\sqrt{32}=7\sqrt{2}-10\sqrt{2}+4\sqrt{2}=\mathbf{\sqrt{2}}$
3. $3\sqrt[3]{16}+2\sqrt[3]{54}=6\sqrt[3]{2}+6\sqrt[3]{2}=\mathbf{12\sqrt[3]{2}}$
4. $4\sqrt[3]{81}-2\sqrt[3]{24}=12\sqrt[3]{3}-4\sqrt[3]{3}=\mathbf{8\sqrt[3]{3}}$
5. $\sqrt{48x^5}=\sqrt{16\cdot 3\cdot x^4\cdot x}=\mathbf{4x^2\sqrt{3x}}$

Part 4 — Simplify Radicals (6–10)

6. $\sqrt[3]{x^{12}y^9}=x^{12/3}\cdot y^{9/3}=\mathbf{x^4y^3}$
7. $\sqrt{a^6b^4}=\mathbf{a^3b^2}$
8. $2\sqrt{45x^5}-x^2\sqrt{20x}=6x^2\sqrt{5x}-2x^2\sqrt{5x}=\mathbf{4x^2\sqrt{5x}}$
9. $3\sqrt{8x^3}-x\sqrt{18x}=6x\sqrt{2x}-3x\sqrt{2x}=\mathbf{3x\sqrt{2x}}$
10. $\sqrt[3]{24a^6b^3}=\sqrt[3]{8\cdot 3\cdot a^6\cdot b^3}=\mathbf{2a^2b\sqrt[3]{3}}$

Part 5 — Extra Challenge (1–5)

1. $5\sqrt{18x^3}+2x\sqrt{50x}-\sqrt{2x^3}=15x\sqrt{2x}+10x\sqrt{2x}-x\sqrt{2x}=\mathbf{24x\sqrt{2x}}$
2. $\sqrt[3]{54x^4y^7}=\sqrt[3]{27\cdot 2\cdot x^3\cdot x\cdot y^6\cdot y}=\mathbf{3xy^2\sqrt[3]{2xy}}$
3. $2\sqrt[3]{40}-3\sqrt[3]{135}=4\sqrt[3]{5}-9\sqrt[3]{5}=\mathbf{-5\sqrt[3]{5}}$
4. $\sqrt[3]{16a^7b^4}+2ab\sqrt[3]{2a^4b}=2a^2b\sqrt[3]{2ab}+2a^2b\sqrt[3]{2ab}=\mathbf{4a^2b\sqrt[3]{2ab}}$
5. $\dfrac{\sqrt{72a^5b^3}}{\sqrt{2ab}}=\sqrt{36a^4b^2}=\mathbf{6a^2b}$

Part 5 — Extra Challenge (6–10)

6. $\sqrt[3]{\dfrac{x^9y^3}{8z^6}}=\mathbf{\dfrac{x^3y}{2z^2}}$
7. $4\sqrt{75x^3}-x\sqrt{48x}+\sqrt{27x^3}=20x\sqrt{3x}-4x\sqrt{3x}+3x\sqrt{3x}=\mathbf{19x\sqrt{3x}}$
8. $\sqrt[3]{x^5y^4}\cdot\sqrt[3]{x^4y^2}=\sqrt[3]{x^9y^6}=\mathbf{x^3y^2}$
9. $3\sqrt{a^3b^5}-ab\sqrt{ab^3}=3ab^2\sqrt{ab}-ab^2\sqrt{ab}=\mathbf{2ab^2\sqrt{ab}}$
10. $\sqrt[3]{250}-\sqrt[3]{2}+\sqrt[3]{16}=5\sqrt[3]{2}-\sqrt[3]{2}+2\sqrt[3]{2}=\mathbf{6\sqrt[3]{2}}$

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